

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
3	(a)			Indicative content: 1. Heat energy from the room will transfer from high temperature to low temperature. 2. The bigger the temperature (gradient) difference the greater the rate of heat flow will be greater. 3. The cavity board is made from foam which is itself a good insulator/bad conductor, which reduces heat loss by conduction. 4. Trapped air in the foam & between the board and the walls also reduces convection currents in the cavity and reduces heat loss by convection. 5. The foam has an inner shiny silver surface that faces into the house and is a poor absorber. Radiation from the house will be reflected back into the house. 6. The outer shiny surface is a poor emitter of i-r radiation. 7. Consequently the temperature of the outer wall / surrounding air is less, reducing convection losses from the outside wall. 5 – 6 marks Expect at least 4 statements. <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i> 3 – 4 marks Expect at least 2 statements Alternatively, a brief description of all three heat loss reduction processes. <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i>	3	3		6		

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					AO1	AO2	AO3	Total	Maths	Prac
				1-2 marks Any correct statement. <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i> 0 marks <i>No attempt made or no response worthy of credit.</i>						
	(b)	(i)		Use of £385 and £110 (1) [or by implication] Payback time = 3.5 years so disagree [or equiv. e.g. 'no'] (1) Alternative: $110 + 110 + 110$ [or 3×110] = 330 (1) < 385 so disagree. (1)		1	1	2	1	
		(ii)		Answers in the table: CO ₂ saving = 640 (1) Carbon footprint = 2 000 (1)		2		2	2	
		(iii)		Mid-terraced house (1) cao <u>lowest</u> {carbon <u>footprint</u> / <u>emission</u> of CO ₂ } (1) NB reference to 'saving' → 1st mark only NB 2nd mark can only be awarded if first mark is given. Do not accept just values from the table unless all three values compared.			2	2		
				Question 3 total	3	6	3	12	3	0